

AP Physics 1: Algebra-Based The Complete Exam Guide

All 8 units explained, including fluids, the four free-response question types, a lab-design playbook, and 10 hard practice questions.

Hi, I'm **Pigo!** I'll drop tips all through this guide.



3 hrs

85 min multiple choice
+ 95 min free response

42+4

multiple-choice +
free-response questions

8 units

kinematics to
fluids

67–68%

scored 3 or higher
in recent years

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01 THE EXAM AT A GLANCE

Two sections, one score

Your AP score (1–5) combines a multiple-choice section and a free-response section, each worth half. A calculator and an equation sheet are allowed on the whole exam.

50%

Section I • Multiple choice • 42 questions

50%

Section II • Free response • 4 questions

Exam structure

Section	Weight & timing	What you do
I • Multiple choice	50% of score 1 hr 25 min 42 questions	Single questions and short sets built around a stimulus such as a diagram, graph or data table. Calculator allowed.
II • Free response	50% of score 1 hr 35 min 4 questions	Four question types: Mathematical Routines • Translation Between Representations • Experimental Design and Analysis • Qualitative/Quantitative Translation.

HYBRID DIGITAL

Click in Bluebook, write on paper

You answer multiple-choice questions and read the free-response questions in the **Bluebook** app, then handwrite your free-response answers in a paper booklet. Practice writing clear, organized work by hand.

PROVIDED ON EXAM DAY

Equation sheet + calculator

A table of equations and constants is provided, and a calculator is allowed throughout. Knowing **when** an equation applies, and its conditions, matters more than memorizing it.

How to use this guide

Just starting?

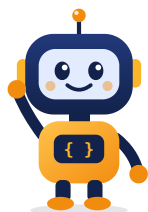
Read pages 2–6, then follow the roadmap on page 8 month by month.

Mid-year?

Check the traps on page 7 and drill the free-response playbook on page 6.

Exam in weeks?

Take the 10 questions on page 9 in 22 minutes, then use the final-four-weeks plan.



Pigo's tip: Physics answers are scored on **reasoning**. Start every explanation from a principle, such as Newton's second law or conservation of energy, then apply it to the situation.

SECTION I

02 The multiple-choice section

42 questions in 85 minutes gives you about 2 minutes per question. Many questions test reasoning with symbols and graphs rather than plugging in numbers.

42

questions, some in stimulus-based sets

~2:00

minutes per question on average

50%

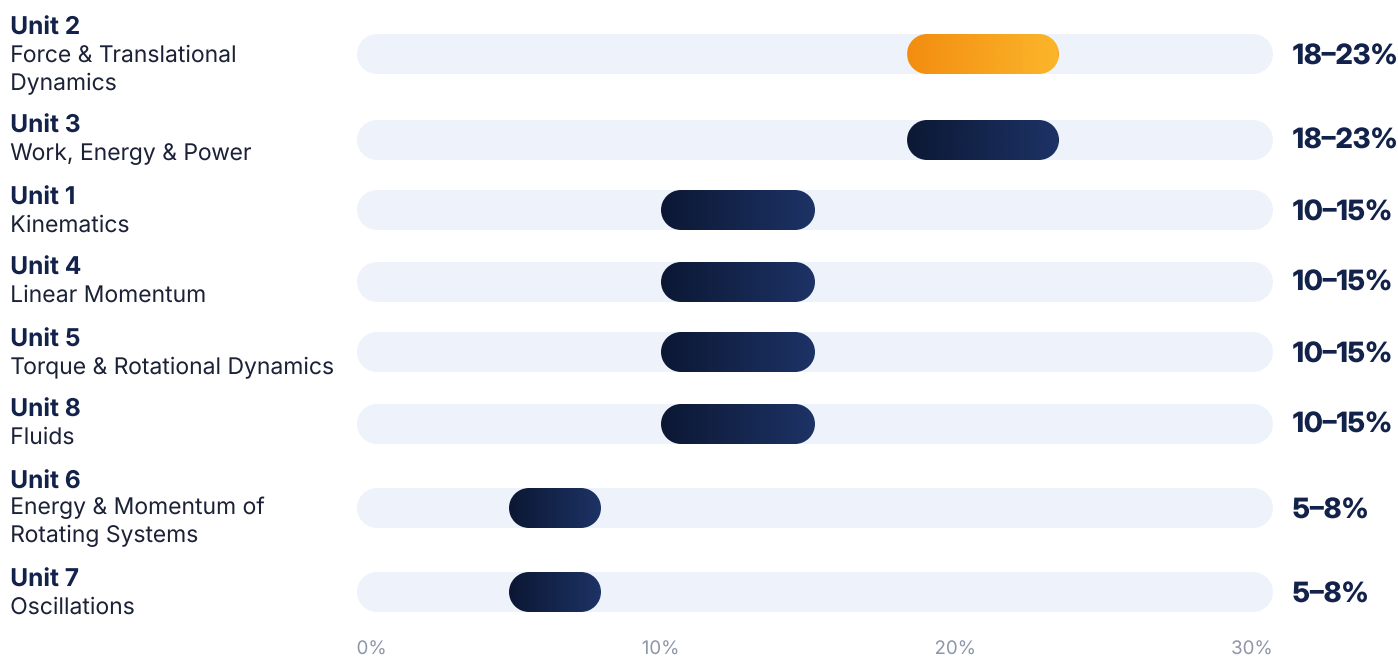
of your total AP score

0

points lost for a wrong answer, so never leave one blank

How the questions are weighted by unit

College Board gives a range for each unit's share of the multiple-choice section. Bars show minimum to maximum.



What this means for your study time: Forces and energy together are about 40% of the section, and they reappear in rotation, oscillations and fluids. Master them first.

Strategies that work on this section

- **Draw a diagram** for every question: forces, fields, circuits or rays. It takes seconds and prevents sign errors.
- **Check limiting cases.** What happens if a mass is zero or an angle is 90° ? Wrong choices often fail this test.
- **Use units to eliminate** answer choices with the wrong dimensions.
- **Read graphs as relationships:** slope and area under the curve both have physical meaning.
- **Work symbolically first**, then plug in numbers only if needed.
- **Flag and move on** after 3 minutes. Return once the quick points are banked.

03 WHAT THE COURSE COVERS

The 8 units, in plain English

Everything on the exam fits into these units.

UNIT 1 10–15%	Kinematics Be ready to: use motion graphs (slope and area); apply kinematic equations in 1D and 2D; solve projectile problems; use reference frames.	UNIT 2 18–23%	Force & Translational Dynamics Be ready to: draw free-body diagrams; apply Newton's laws to systems; handle friction, springs, inclines and pulleys; analyze circular motion and gravitation.
UNIT 3 18–23%	Work, Energy & Power Be ready to: calculate work from force and graphs; use the work-energy theorem; apply conservation of energy with energy bar charts; calculate power.	UNIT 4 10–15%	Linear Momentum Be ready to: use impulse and force-time graphs; apply conservation of momentum; classify elastic and inelastic collisions; track the center of mass.
UNIT 5 10–15%	Torque & Rotational Dynamics Be ready to: calculate torque; use rotational kinematics; apply $\Sigma \tau = I\alpha$; analyze static equilibrium and rotational inertia.	UNIT 6 5–8%	Energy & Momentum of Rotating Systems Be ready to: use rotational kinetic energy; analyze rolling without slipping; apply conservation of angular momentum.
UNIT 7 5–8%	Oscillations Be ready to: describe simple harmonic motion; find periods of springs and pendulums; track energy in oscillators.	UNIT 8 10–15%	Fluids Be ready to: use density and pressure; apply buoyancy (Archimedes); use the continuity equation and Bernoulli's equation.

Algebra-based, not calculus-based. You won't need derivatives or integrals, but you must manipulate symbols confidently and read slopes and areas from graphs.

SECTION II

04

The 4 free-response questions

95 minutes, 4 questions. The four question types are the same every year, so you can practice each on purpose.

MR

TBR

LAB

QQT

Mathematical Routines and Lab questions are typically worth the most points

MR

Mathematical Routines

Derive and calculate

Use equations and principles to derive expressions and calculate quantities, often starting symbolically.

Start from a fundamental equation and show each substitution.

TBR

Translation Between Representations

Diagrams, graphs, equations

Draw a free-body diagram, energy bar chart or graph, then write equations that are consistent with it.

Arrows on free-body diagrams start on the object and are labeled.

LAB

Experimental Design & Analysis

Plan and analyze

Design a procedure, choose what to graph to get a straight line (linearize), then use the slope to find a quantity.

Linearize: graph the quantities that make $y = mx + b$ a straight line.

QQT

Qualitative/Quantitative Translation

Explain, then verify

Explain a situation in words with physics reasoning, then derive an equation that agrees with your explanation.

Make sure your words and your equation predict the same trend.

◆ A time plan for the 95 minutes

MR

~25 min

TBR

~22 min

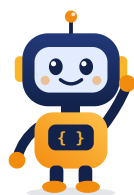
LAB

~25 min

QQT

~23 min

A suggested split, not a rule. Start with the question type you're strongest at, and never leave a question blank.



Points are given for **derivations**. "Plugging in" without a starting equation often loses credit.

FREE-RESPONSE PLAYBOOK

05

Anatomy of a full-credit answer

Here is a short practice prompt with a full-credit answer, and the habits that earn points.

PRACTICE TASK • WRITTEN BY ACT SAT GO

Practice prompt (QQT style): Block A (mass m) moves at speed v_o on a frictionless surface and collides with block B (mass $2m$) at rest. They stick together. (a) Without equations, explain whether kinetic energy is conserved. (b) Derive the final speed in terms of v_o . (c) Show that your answer to (b) agrees with (a).

Full-credit response

① (a) Kinetic energy is **not** conserved. The blocks stick together, so some kinetic energy becomes internal energy (deformation and heat). Momentum is conserved because no net external horizontal force acts.

② (b) $m v_o = (m + 2m) v_f$, so $v_f = v_o/3$.

③ (c) $K_i = \frac{1}{2}mv_o^2$. $K_f = \frac{1}{2}(3m)(v_o/3)^2 = (1/6)mv_o^2 = \frac{1}{3}K_i$.

④ $K_f < K_i$, which matches the loss of kinetic energy described in (a).

- ① **Name the principle** and the reason it applies.
- ② **Start from conservation of momentum** and solve symbolically.
- ③ **Compute both energies** from your result.
- ④ **Connect back** so the words and the math agree.

Five habits for every free-response question

1

Answer every lettered part, in order, with a label.

2

Show the equation, the substitution and the units.

3

Justify with a cause → effect sentence, not a word.

4

Use the reference sheet; you don't need to memorize it.

5

Blank part? Write the relationship you would use. It can earn credit.

06 LEARN FROM OTHERS

10 traps that cost points

Meet **Glitch**. He lives in these ten mistakes. Squash each one before May.

Normal force always equals **mg**! (It doesn't.)



01 Normal force = mg

Only on a flat surface with no other vertical forces and no vertical acceleration. On an incline it is $mg \cos \theta$.

02 Force in the direction of motion

A moving object needs no forward force. Net force tells you acceleration, not velocity.

03 Centripetal force as a new force

Never draw " F_c " on a free-body diagram. Tension, gravity or friction provides it.

04 Momentum vs energy

In every collision without external forces, momentum is conserved. Kinetic energy is conserved only in elastic collisions.

05 Sign errors

Choose a positive direction at the start and stick to it for velocity, acceleration and force.

06 Mass and pendulum period

A pendulum's period doesn't depend on mass or (for small angles) amplitude: $T = 2\pi\sqrt{L/g}$.

07 Rolling objects

Rolling objects share energy between translation and rotation, so they move slower than sliding frictionless ones.

08 Buoyant force

$F_b = \rho_{\text{fluid}} V_{\text{displaced}} g$. Use the fluid's density, not the object's.

09 Faster fluid, higher pressure

In Bernoulli flow, where the fluid moves faster the pressure is lower.

10 Words and math disagree

In QQT questions, an explanation that contradicts your equation loses points on both.

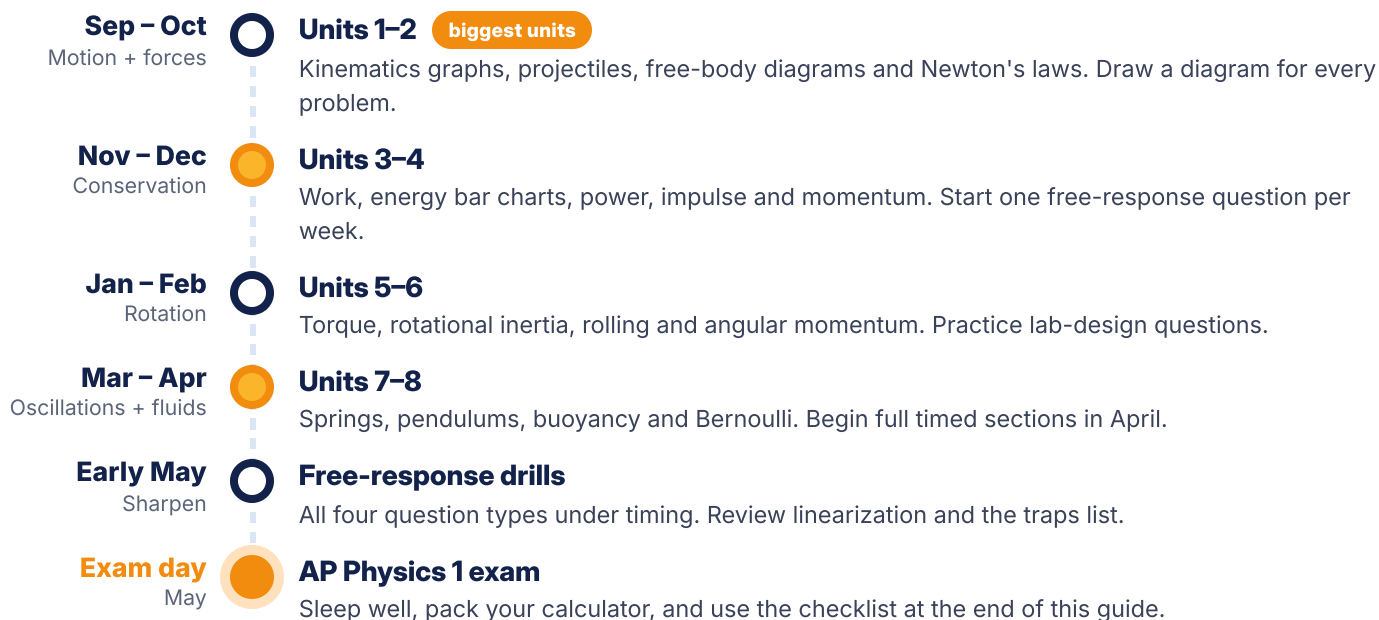
Quick self-check: a ball is thrown straight up. At the top, what are its velocity and acceleration? (Velocity 0; acceleration 9.8 m/s^2 downward, since gravity never switches off.)

07

PLAN YOUR YEAR

Your study roadmap to May

A steady plan beats a last-minute rush. Shift the months to match your teacher's pacing, and keep doing a few problems from older units every week.



◆ The final four weeks

Week	Focus	Target
4 weeks out	Diagnose: 21 multiple-choice questions plus 2 free-response questions	List your two weakest units
3 weeks out	Fix weak units; one of each free-response type	80%+ on unit drills
2 weeks out	Full 3-hour practice exam	Finish both sections on time
Final week	Light review: equation sheet, diagram rules, traps list	Rest the night before

08 TEST YOURSELF 10 practice questions

Original questions written by ACT SAT GO in the style of the exam, pitched at the harder end. Use $g = 9.8 \text{ m/s}^2$ unless stated. Give yourself **22 minutes** for all ten, then check page 10.

1 UNIT 1 • PROJECTILES

A ball is launched from level ground at 20 m/s and 30° above horizontal. Ignoring air resistance, about how far away does it land?

- A 17.7 m
- B 20.4 m
- C 35.3 m
- D 40.8 m

2 UNIT 2 • CONNECTED MASSES

Masses of 5 kg and 3 kg hang on opposite sides of a light, frictionless pulley. What is the tension in the string?

- A 19.6 N
- B 29.4 N
- C 36.8 N
- D 49.0 N

3 UNIT 2 • CIRCULAR MOTION

A car drives over the top of a hill shaped like a circle of radius 40 m. What is the greatest speed at which its tires stay in contact with the road?

- A 14 m/s
- B 20 m/s
- C 28 m/s
- D 40 m/s

4 UNIT 4 • COLLISIONS

A 2 kg cart at 3 m/s collides with and sticks to a 1 kg cart at rest. How much kinetic energy is lost?

- A 0 J
- B 3 J
- C 6 J
- D 9 J

5 UNIT 6 • ROLLING

A solid cylinder ($I = \frac{1}{2}MR^2$) rolls without slipping from rest down a ramp, dropping a vertical height of 1.2 m. What is its speed at the bottom?

- A 3.4 m/s
- B 4.0 m/s
- C 4.8 m/s
- D 5.6 m/s

6 UNIT 5 • TORQUE

A uniform 4 m beam of mass 20 kg is supported at one end and by a vertical cable at the other. A 30 kg child stands 1 m from the cable end. What is the tension in the cable?

- A 196 N
- B 245 N
- C 319 N
- D 490 N

7 UNIT 7 • OSCILLATIONS

A pendulum on Earth has a period of 2.0 s. On a planet where g is four times larger, the same pendulum's period is:

- A 0.5 s
- B 1.0 s
- C 2.0 s
- D 4.0 s

8 UNIT 3 • SPRING ENERGY

A 0.5 kg block is pressed against a spring ($k = 200 \text{ N/m}$), compressing it 0.10 m, then released on a frictionless surface. What is the block's speed after leaving the spring?

- A 1.0 m/s
- B 1.4 m/s
- C 2.0 m/s
- D 4.0 m/s

9 UNIT 8 • FLUIDS

Water flows through a horizontal pipe that narrows from area A to area $A/2$. Compared with the wide section, the narrow section has:

- A Higher speed and lower pressure
- B Higher speed and higher pressure
- C Lower speed and higher pressure
- D The same speed and lower pressure

10 UNIT 8 • BUOYANCY

A wooden block of density 600 kg/m^3 floats in fresh water ($1,000 \text{ kg/m}^3$). What fraction of its volume is **below** the waterline?

- A 0.40
- B 0.60
- C 0.67
- D 1.67

09

CHECK YOUR WORK

Answer key & explanations

Read every explanation, even for questions you got right. Each one points to a pattern that keeps coming back on the exam.

#	Answer	Why
1	C • 35.3 m	$R = v_o^2 \sin(2\theta)/g = 400 \times \sin 60^\circ/9.8 \approx 35.3$ m. Choice D drops the $\sin 2\theta$ factor.
2	C • 36.8 N	$a = (5 - 3)g/8 = 2.45$ m/s ² . For the 3 kg mass: $T - 3g = 3a$, so $T = 3(9.8 + 2.45) \approx 36.8$ N. Tension lies between the two weights.
3	B • 20 m/s	At the top, contact is lost when gravity alone provides the centripetal force: $mg = mv^2/r$, so $v = \sqrt{gr} = \sqrt{392} \approx 20$ m/s.
4	B • 3 J	Momentum: $2 \times 3 = 3v$, so $v = 2$ m/s. KE before = 9 J, after = $\frac{1}{2}(3)(4) = 6$ J. Lost: 3 J.
5	B • 4.0 m/s	$mgh = \frac{1}{2}mv^2 + \frac{1}{2}(\frac{1}{2}mR^2)(v/R)^2 = \frac{3}{4}mv^2$. $v = \sqrt{(4gh/3)} = \sqrt{15.7} \approx 4.0$ m/s.
6	C • 319 N	Torques about the support: $T(4) = 20g(2) + 30g(3) = 130g$. $T = 32.5 \times 9.8 \approx 319$ N.
7	B • 1.0 s	$T = 2\pi\sqrt{L/g}$. Four times g halves the period: 2.0 s $\rightarrow$ 1.0 s.
8	C • 2.0 m/s	Spring energy $\frac{1}{2}kx^2 = \frac{1}{2}(200)(0.01) = 1$ J = $\frac{1}{2}(0.5)v^2$, so $v = 2.0$ m/s.
9	A	Continuity (Av constant): halving the area doubles the speed. Bernoulli: faster flow at the same height means lower pressure.
10	B • 0.60	Floating: weight = buoyant force, so $\rho_{\text{block}}V = \rho_{\text{water}}V_{\text{sub}}$. $V_{\text{sub}}/V = 600/1000 = 0.60$.

9–10

Strong. Move on to full timed free-response practice.

6–8

On track. Revisit the unit behind every miss on page 4.

0–5

Build foundations now. Drill free-body diagrams and energy conservation (Units 2–3) first.

10 KNOW THE TARGET

How students score

In recent years, about 67–68% of students earned a 3 or higher on AP Physics 1, and about 19–20% earned a 5. Steady preparation is what moves a score up a full point.

67–68%

of students earned a 3 or higher in recent years

19–20%

earned the top score of 5 in recent years

3+

is the score many colleges accept for credit or placement

What each AP score means

5

Extremely well qualified

4

Very well qualified

3

Qualified

2

Possibly qualified

1

No recommendation

Ranges reflect recent College Board score distributions and shift a little each year. Credit and placement policies vary by college, so check each college's AP policy.

Exam-day checklist

- ☐ **Bluebook** is installed and updated, and your device passed the exam check
- ☐ Device **fully charged**, plus **two pencils and a dark-ink pen** for the free-response booklet
- ☐ Admission ticket and photo ID, as your school requires
- ☐ Confirm your **exam day and start time** in May with your AP coordinator
- ☐ An **approved calculator** with fresh batteries
- ☐ Plan your time: about **2 min** per multiple-choice question and **23 min** per free-response question
- ☐ Answer **every** multiple-choice question. Wrong answers don't lose points
- ☐ A **ruler or straightedge** for graphs and diagrams, if permitted

In the free-response section: start from a fundamental principle, draw clean labeled diagrams, keep words and equations consistent, and show every step of a derivation before plugging in numbers.



ACT SAT GO • 1-ON-1 AP TUTORING

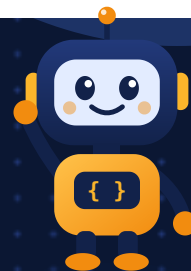
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